

Laboratory assignment

Component 1

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1 Overview of the JADE Framework

JADE (Java Agent DEvelopment Framework) [BCG07] is a well-known framework for creating multi-agent systems, whose architectures can simulate intelligent, adaptive behaviours by allowing agents to communicate and learn together. It provides a robust framework for building multi-agent systems compliant with FIPA (Foundation for Intelligent Physical Agents) standards, supporting the creation of complex, interoperable agent-based applications, particularly for environments requiring distributed, intelligent behaviour. JADE uses the Agent-Oriented Programming software paradigm as a main component of its framework, which is one of the most utilized in the scope of building complex systems complying with the multi-agent system specification. JADE simplifies the implementation of multi-agent systems through a middleware that complies with the FIPA specifications for interoperable intelligent multi-agent systems. Although JADE is sometimes considered a bit outdated, it remains highly relevant because it provides well-structured and intuitive documentation that allows developers to grasp the core concepts of agency quickly.

2 Programming Language

JADE is written in Java, so it is supplied with a vast array of third-party libraries that can be easily integrated into any agent-based project. This choice of language ensures a high level of ease in developing multi-agent system applications, even for those with a basic knowledge of agent theory, as it leverages standard object-oriented principles.

3 Agent Design

In terms of architectural design, the class Agent is a common superclass for all user-defined software agents. Every agent is essentially a computational process inside the Agent Platform (AP) that offers one or more services, which are typically published in a service descriptor for discovery.

The lifecycle of these agents is managed by the framework, starting from a setup phase where the agent initializes its internal state and registers with the directory services. Because JADE agents are designed as autonomous entities, they encapsulate their own state and behavior, interacting with the platform through a set of predefined methods that handle transitions.

4 Concurrent and Parallel Execution

One of the most powerful features of the framework is that an agent can execute several behaviours concurrently. In JADE, an agent is a single-threaded entity to avoid the complexities of traditional concurrent programming, such as deadlocks or race conditions. However, it achieves multi-tasking by using a scheduler that manages a queue of behaviors. These behaviors are executed cooperatively, meaning the agent cycles through its active tasks and gives each one a chance to run. This allows a single agent to handle multiple responsibilities, such as responding to messages, monitoring environment variables, or performing scheduled calculations, without the overhead of managing multiple system threads.

5 Agent Communication

Communication in JADE is based on asynchronous message passing, which allows agents to stay autonomous because they do not have to stop and wait for replies. Every agent has a private message queue where the JADE runtime delivers incoming messages. These messages follow the FIPA-ACL standard and include specific fields like the sender, the list of receivers, and the "performative," which explains the intent of the message (such as a request or an answer). To ensure these messages move reliably across different machines, JADE uses a Message Transport Service that handles the technical delivery via protocols like HTTP or TCP/IP. The framework also supports ontologies, which act as shared vocabularies so that agents can correctly interpret the data they receive. By using a common dictionary, agents can understand the context of a conversation, ensuring that the information is not just delivered, but also understood by both parties.

References

- [BCG07] Fabio Bellifemine, Giovanni Caire, and Dominic Greenwood. *Developing Multi-Agent Systems with JADE*. Wiley, February 2007.